

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and
Commerce Department of Information Technology (B.Sc.I.T Semester VI)
Business Analytics
Practical I

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Class : TYIT	Date/Time of Submission : 31/07/2026

```
import pandas as pd
df = pd.read_csv("supermarket_sales.csv")
print(df.head())
```

```
...
```

	Invoice ID	Branch	City	Customer type	Gender	\
0	750-67-8428	A	Yangon	Member	Female	
1	226-31-3081	C	Naypyitaw	Normal	Female	
2	631-41-3108	A	Yangon	Normal	Male	
3	123-19-1176	A	Yangon	Member	Male	
4	373-73-7910	A	Yangon	Normal	Male	

	Product line	Unit price	Quantity	Tax 5%	Total	Date	\
0	Health and beauty	74.69	7	26.1415	548.9715	1-5-2019	
1	Electronic accessories	15.28	5	3.8200	80.2200	3-8-2019	
2	Home and lifestyle	46.33	7	16.2155	340.5255	3-3-2019	
3	Health and beauty	58.22	8	23.2880	489.0480	1-27-2019	
4	Sports and travel	86.31	7	30.2085	634.3785	2-8-2019	

	Time	Payment	cogs	gross margin percentage	gross income	Rating
0	13:08	Ewallet	522.83	4.761905	26.1415	9.1
1	10:29	Cash	76.40	4.761905	3.8200	9.6
2	13:23	Credit card	324.31	4.761905	16.2155	7.4
3	20:33	Ewallet	465.76	4.761905	23.2880	8.4
4	10:37	Ewallet	604.17	4.761905	30.2085	5.3

<pre>sales = df["Tax 5%"]</pre>	<pre>...</pre>
<pre>print("Mean :", sales.mean())</pre>	<pre>Mode :</pre>
Mean : 15.379368999999999	0 4.1540
<pre>print("Median :", sales.median())</pre>	1 4.4640
Median : 12.088000000000001	2 8.3770
<pre>print("Mode :")</pre>	3 9.0045
<pre>print(sales.mode())</pre>	4 10.3260
	5 10.3635
	6 12.5700
	7 13.1880
	8 22.4280
	9 39.4800
	Name: Tax 5%, dtype: float64

```

print("Minimum :", sales.min())
... Minimum : 0.5085

print("Maximum :", sales.max())
... Maximum : 49.65

print("Range :", sales.max() - sales.min())
Range : 49.1415

print("Variance :", sales.std())
Variance : 11.708825480998659

print("Standard Deviation :", sales.std())
Standard Deviation : 11.708825480998659

print("Count :", sales.count())
Count : 1000

print("Sum :", sales.sum())
Sum : 15379.368999999999

print("\nDescriptive Statistics")
print(sales.describe())

Descriptive Statistics
count    1000.000000
mean      15.379369
std       11.708825
min        0.508500
25%        5.924875
50%       12.088000
75%       22.445250
max       49.650000
Name: Tax 5%, dtype: float64

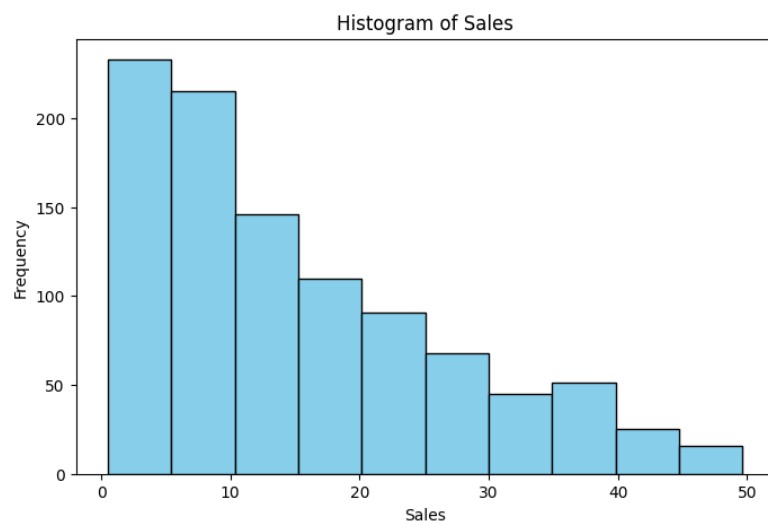
```

Practical-2(B)

```

import matplotlib.pyplot as plt
import seaborn as sns
plt.figure(figsize=(8,5))
plt.hist(sales, bins=10, color="skyblue", edgecolor="black")
plt.title("Histogram of Sales")
plt.xlabel("Sales")
plt.ylabel("Frequency")
plt.show()

```



```
plt.figure(figsize=(8,3))
```

```
sns.boxplot(x=sales, color="orange")  
plt.title("Box Plot of Sales")  
plt.xlabel("Sales")  
plt.show()
```

